Yuezhe **Yang**

Bean-Young · Uyzyang · bean-young.github.io · (+86) 133-4595-9676 · yangyuezhe@gmail.com
Fengshan Street, Yuyao, Ningbo, Zhejiang, China · Aug 2004

Bio. I am a PhD student at USyd School of CS and a research intern at ITM, SJTU. I received a B.Eng. in AI from AHU and previously interned at UoA and the International Brain Science and Engineering Research Center.

Research interests. My research focuses on **Medical AI**, including **3D reconstruction** for ultrasound, CT, MRI, and PET ; **intelligent diagnosis** with large language models, agents, reinforcement learning, and data-centric learning ; and **trustworthy AI** for domain shift, safety, and clinical translation.

Education

Sep 2022 – Jun 2026

Doctor of Philosophy, The University of Sydney, Sydney
Major in **Computer Science** | *Advisor : Prof. Jinman Kim*
Bachelor of Engineering, Anhui University, Hefei
Major in **Artificial Intelligence** | *Advisor : Prof. Zhe Jin*
Score : 90.14/100 | **GPA** : 3.95/5.0 | **Rank** : 9/262

Publications

∗ Co-first author ; † Corresponding author.

- **Yang Y***, Chen Y*, Dong X†, Zhang J, Long C, Jin Z & Dai Y. An annotated heterogeneous ultrasound database. *Scientific Data* (2025). **JCR Q1**.
- Zhu K*, **Yang Y***, Chen Y*,†, Feng R, Chen D, Fan B, Liu N, Li Y & Wang X. EM-Net : Effective and morphology-aware network for skin lesion segmentation. *Expert Systems with Applications* (2025). **JCR Q1**.
- **Yang Y**, Ruan Q, Cai W, Dong Y, Yang D, Dong X†, Jin Z & Dai Y. UltraGS : Real-Time Physically-Decoupled Gaussian Splatting for Ultrasound Novel View Synthesis. *IEEE International Conference on Multimedia & Expo* (2026). **CORE A**.
- Dong Y*, Liu M*, Feng J*, **Yang Y***, Dai Y & Jin Z†. Federated Learning-Based Virtual Dual-Energy CT Generation from Single-Energy CT for Gout Detection. *Digital Health* (2025). **JCR Q1**.
- Dong X†, Liu Z, **Yang Y**, Lai Y-L & Jin Z. Conductor : Dynamically orchestrating pipeline parallelism with multi-granularity control. *Future Generation Computer Systems* (2027). **JCR Q1**.
- Dong X, **Yang Y**, Lü X, Wang L, Zhang H, Chen Y, Zhang D & Jin Z. PET image reconstruction method based on prior image and PET image 3D perception method. *Chinese Invention Patent* (2024).
- **Yang Y**, Dong X, Cai W, Yang D & Jin Z. 3D Gaussian model generation and rendering method for low-dose PET image processing. *Chinese Invention Patent* (2025).

Experiments

Dec. 2023 Mar. 2025	International Brain Science and Engineering Research Center, Anhui University, Research Intern **Supervisor :** Prof. Xingbo Dong. **Contributions :** Developed domain-generalizable reconstruction and diagnosis methods for ultrasound, PET, and skin imaging ; co-authored published and accepted papers and led *DeepPET*, which received national and provincial innovation awards.
Jul. 2025 Oct. 2025	Vision and Learning Lab, University of Alberta, Research Intern **Supervisor :** Prof. Li Cheng. **Contributions :** Developed physics-informed deep learning for 3D CT reconstruction in low-dose, limited-angle, and high-noise settings, improving image quality under constrained acquisition.
Dec. 2025 Oct. 2026	Institute of Translational Medicine, Shanghai Jiao Tong University, Research Intern **Supervisor :** Prof. Lei Bi. **Contributions :** Develop multimodal clinical intelligence with vision–language models and 3D medical reconstruction to support clinically translatable AI.

Honors

- 2026 : **Outstanding Graduate of Anhui Province** *(Top 3% provincewide)*
- 2024 : **The National Scholarship** *(Top 0.4%, CNY 10,000)*
- 2023–2025 : **Undergraduate Innovation Training Project** *(Outstanding completion, CNY 6,000)*
- 2025 : **Undergraduate CSC Government Scholarship** *(30 recipients, CAD 5,400)*
- 2024 : **Anhui Province Student Innovation Competition Gold Award** *(Higher Education Track, project lead)*